

To Design and Verify a Low-Power 5T OTA in GF180MCU Technology That Meets Gain, GBW, Power, Output Common-Mode Voltage, and Saturation Requirements Using the gm/ID Design Methodology.

Technology

GF180MCU PDK

TT Corner

Temperature = 27°C

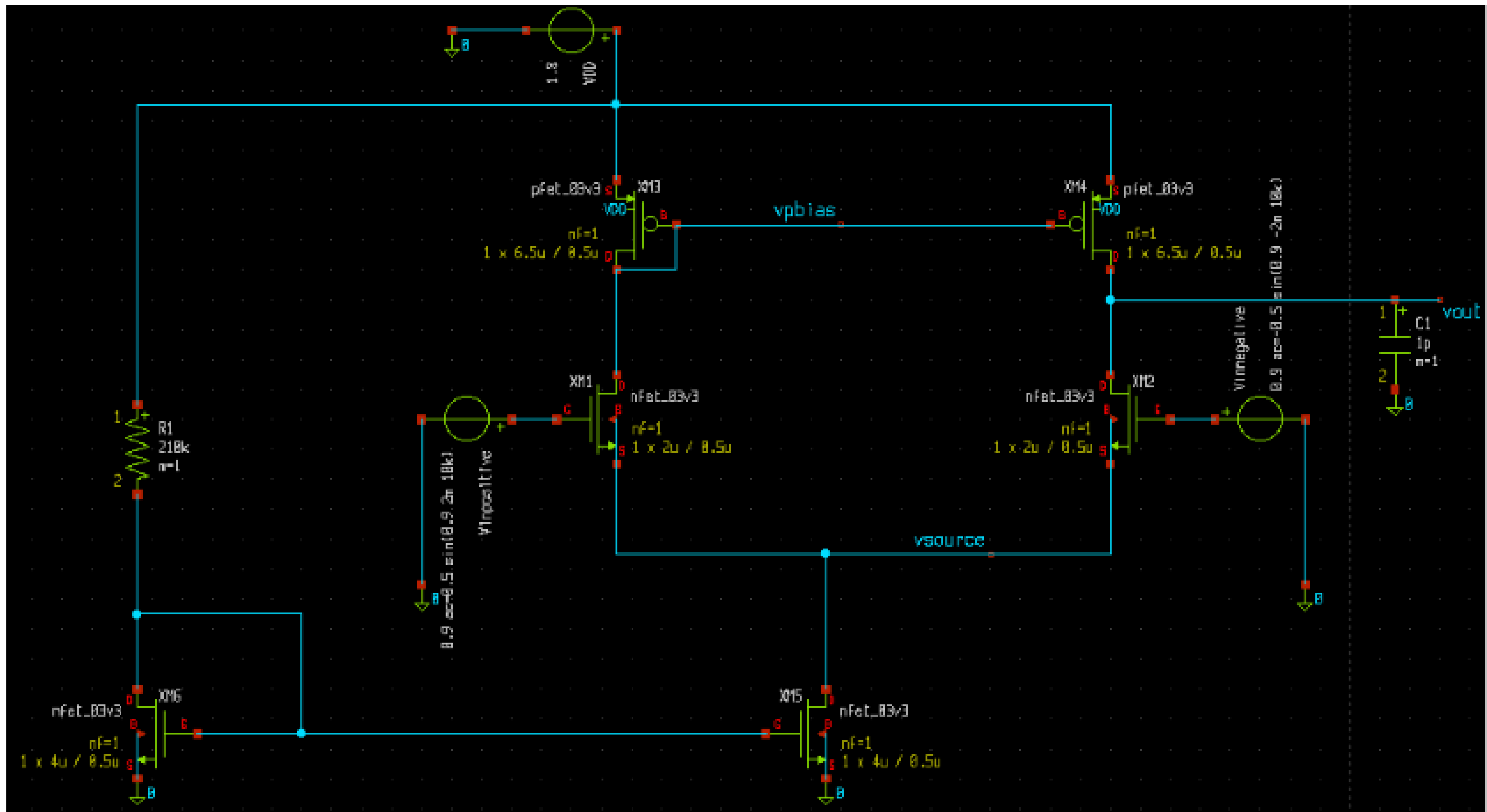
Supply

VDD = 1.8 V

Specifications

- DC Gain ≥ 35 dB
- GBW ≥ 5 MHz
- Power ≤ 50 μ W
- CL = 1 pF
- Output Common Mode: 0.8 V – 1.2 V
- All MOSFETs in Saturation

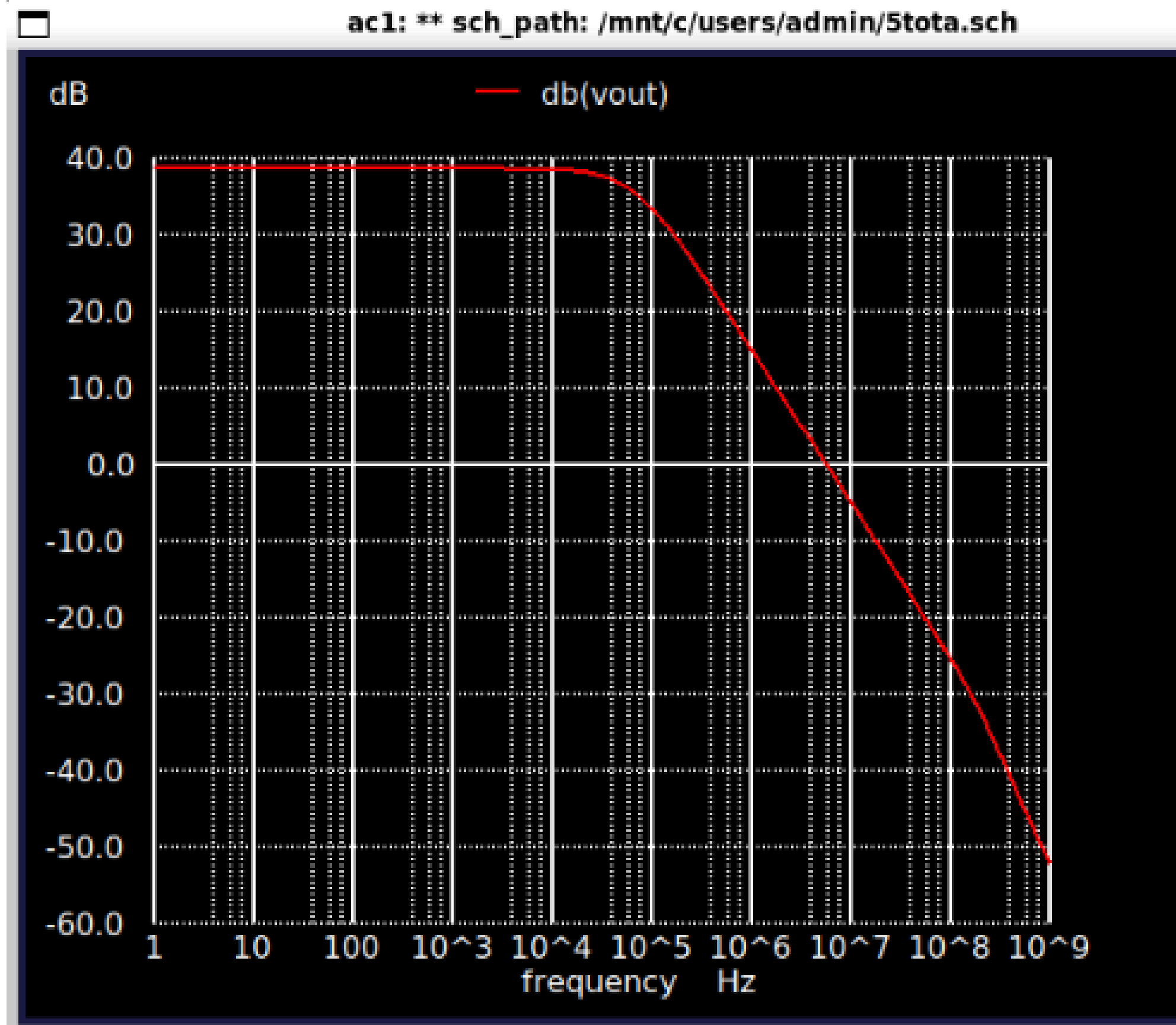
Schematic



Operating Point Verification

```
5TOTA.spice" -a || sh
ngspice 2 -> print all\
@m.xm1.n0[id] = 2.265055e-06
@m.xm1.n0[vds] = 9.345039e-01
@m.xm1.n0[vdsat] = 1.036069e-01
@m.xm2.n0[id] = 2.265055e-06
@m.xm2.n0[vds] = 9.345039e-01
@m.xm2.n0[vdsat] = 1.036069e-01
@m.xm3.n0[vds] = 7.042916e-01
@m.xm3.n0[vdsat] = 9.765604e-02
@m.xm4.n0[vds] = 7.042916e-01
@m.xm4.n0[vdsat] = 9.765604e-02
@m.xm5.n0[id] = 4.530111e-06
@m.xm5.n0[vds] = 1.611979e-01
@m.xm5.n0[vdsat] = 1.056780e-01
@m.xm6.n0[id] = 5.016229e-06
@m.xm6.n0[vds] = 7.465885e-01
@m.xm6.n0[vdsat] = 1.060783e-01
net1 = 1.800000e+00
net2 = 9.000000e-01
net3 = 9.000000e-01
net4 = 7.465917e-01
vdd = 1.515752e+00
vdd#branch = -9.54634e-06
vinnegative#branch = 0.000000e+00
```

GBW Verifiatiion



$$A_0 \approx 89$$

$$f(-3\text{dB}) = 58.8 \text{ kHz}$$

$$\text{GBW} \approx 5.2 \text{ MHz}$$

Specification Verification

- ✓ Differential DC Gain = 39 dB (Requirement: ≥ 35 dB)
- ✓ Gain-Bandwidth Product (GBW) = 5.2 MHz (Requirement: ≥ 5 MHz)
- ✓ Power Consumption = 8.15 μ W (Requirement: ≤ 50 μ W)
- ✓ Output Common-Mode Voltage (VOCM) = 1.095 V (Requirement: 0.8 V – 1.2 V)
- ✓ Tail Current = 4.53 μ A (Generated using NMOS current mirror biasing)
- ✓ All MOSFETs Operate in Saturation